Lecture 9:
Convolution, covariance and correlation

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Outline

1. **Review**
2. **Convolution**
3. **Covariance and correlation**
4. **Recitation**
5. **Summary**

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Review

- Discrete case

$$p_Y(y) = \mathbf{P}(g(X) = y) = \sum_{x:\, g(x)=y} p_X(x)$$

- Continuous case
  - Obtain CDF of Y: $F_Y(y) = \mathbf{P}(Y \leq y)$
  - Differentiate to get $f_Y(y) = \frac{dF_Y}{dy}(y)$
- The linear case $Y = aX + b$

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

- The monotonic case

$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|$$

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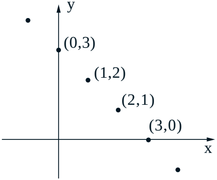
- $W = X + Y$; X, Y independent



$$
\begin{aligned}
p_W(w) &= \mathbf{P}(X+Y=w) \\
&= \sum_{\{(x,y) \mid x+y=w\}} \mathbf{P}(X=x, Y=y) \\
&= \sum_x \mathbf{P}(X=x)\mathbf{P}(Y=w-x) \\
&= \sum_x p_X(x) p_Y(w-x)
\end{aligned}
$$

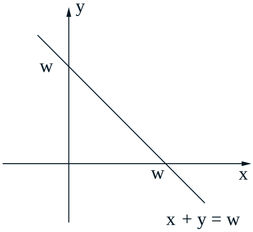
- PMF p_W is called the **convolution** of the PMFs of X and Y.

The distribution of $X + Y$

- Mechanics:
  - Put the PMF's on top of each other
  - Flip the PMF of Y
  - Shift the flipped PMF by w (to the right if $w > 0$)
  - Cross-multiply and add

Continuous case

- $W = X + Y$; X, Y independent



- Conditional CDF: $\mathbf{P}(W \leq w | X = x) = \mathbf{P}(Y \leq w - x)$
- Take derivative w.r.t. w, $f_{W|X}(w \mid x) = f_Y(w - x)$
- Joint PDF: $f_{W,X}(w, x) = f_X(x) f_{W|X}(w \mid x) = f_X(x) f_Y(w - x)$
- $f_W(w) = \int_{-\infty}^{\infty} f_{W,X}(w, x)\, dx = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x)\, dx$

Example

- The RVs X and Y are independent and uniformly distributed in the interval $[0,1]$. The PDF of $Z = X + Y$ is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

- The integrand is equal to 1 for $0 \leq x \leq 1$ and $0 \leq z - x \leq 1$ and is zero otherwise. Thus,

$$f_Z(z) = \begin{cases} \min\{1, z\} - \max\{0, z-1\}, & 0 \leq z \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

The sum of two independent normal r.v.'s

- $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, independent
- Joint PDF

$$
\begin{aligned}
f_{X,Y}(x,y) &= f_X(x) f_Y(y) \\
&= \frac{1}{2\pi\sigma_x\sigma_y} \exp\left\{ -\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-\mu_y)^2}{2\sigma_y^2} \right\}
\end{aligned}
$$

which is constant on the ellipse where

$$
\frac{(x-\mu_x)^2}{2\sigma_x^2} + \frac{(y-\mu_y)^2}{2\sigma_y^2} = \text{Constant}
$$

- Ellipse is a circle when $\sigma_x = \sigma_y$.

The sum of two independent normal r.v.'s

- $X \sim N(0, \sigma_x^2)$, $Y \sim N(0, \sigma_y^2)$, independent
- Let $W = X + Y$

$$\begin{aligned} f_W(w) &= \int_{-\infty}^{\infty} f_X(x) f_Y(w-x)\, dx \\ &= \frac{1}{2\pi\sigma_x\sigma_y} \int_{-\infty}^{\infty} e^{-x^2/2\sigma_x^2} e^{-(w-x)^2/2\sigma_y^2}\, dx \\ \text{(algebra)} &= ce^{-\gamma w^2} \end{aligned}$$

- Conclusion: W is normal.
- Mean $= 0$, variance $= \sigma_x^2 + \sigma_y^2$.
- Same argument for nonzero mean case.

The difference of two independent r.v.'s

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- Consider the case where X and Y are independent exponential RVs with parameter λ. Find the PDF for $X - Y$.
- Fix some $z \geq 0$ and note that $f_Y(x-z)$ is nonzero when $x \geq z$. Thus,

$$\begin{aligned} f_{X-Y}(z) &= \int_{-\infty}^{\infty} f_X(x) f_{-Y}(z-x)\, dx \\ &= \int_{-\infty}^{\infty} f_X(x) f_Y(x-z)\, dx \\ &= \int_z^{\infty} \lambda e^{-\lambda x} \lambda e^{-\lambda(x-z)}\, dx \\ &= \frac{\lambda}{2} e^{-\lambda z}. \end{aligned}$$

- The answer for the case $z < 0$

$$f_{X-Y}(z) = f_{Y-X}(z) = f_{-(X-Y)}(z) = f_{X-Y}(-z)$$

- The first equality holds by symmetry.

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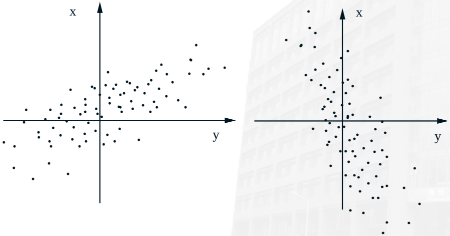
Definition of Covariance

The **covariance** of two RVs X and Y, denoted by $\mathrm{cov}(X,Y)$, is defined by

$$\mathrm{cov}(X,Y) = \mathbf{E}\big[(X-\mathbf{E}[X])(Y-\mathbf{E}[Y])\big].$$

X and Y are **uncorrelated** if $\mathrm{cov}(X,Y)=0$.

- **Zero-mean case:** $\mathrm{cov}(X,Y) = \mathbf{E}\left[XY\right]$



Covariance

- **An alternative formula**:

$$\text{cov}(X,Y) = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y]$$

- **Key properties**: for any RVs X, Y and Z, and any scalars a and b, we have
  - $\text{cov}(X,X) = \text{var}(X)$
  - $\text{cov}(X, aY+b) = a \cdot \text{cov}(X,Y)$
  - $\text{cov}(X, Y+Z) = \text{cov}(X,Y) + \text{cov}(X,Z)$
  - independent $\Longrightarrow$ $\text{cov}(X,Y) = 0$ (**converse is not true**)

Variance of the sum of RVs

- $\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{var}(X_i) + \sum_{\{(i,j)|i\neq j\}} \mathrm{cov}(X_i, X_j)$
- In paticular, if X_1, X_2 are RVs with finite variance, we have

$$\mathrm{var}(X_1 + X_2) = \mathrm{var}(X_1) + \mathrm{var}(X_2) + 2\mathrm{cov}(X_1, X_2).$$

- Denote $\tilde{X}_i = X_i - E[X_i]$. This can be seen from the calculation,

$$\begin{aligned}
\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) &= \mathbf{E}\left[(\sum_{i=1}^{n} \tilde{X}_i)^2\right] \\
&= \sum_{i=1}^{n} \mathbf{E}[\tilde{X}_i^2] + \sum_{\{(i,j)|i\neq j\}} \mathbf{E}[\tilde{X}_i \tilde{X}_j] \\
&= \sum_{i=1}^{n} \mathrm{var}(X_i) + \sum_{\{(i,j)|i\neq j\}} \mathrm{cov}(X_i, X_j)
\end{aligned}$$

Example

- Consider the hat problem again, where n people throw their hats in a box and then pick a hat at random. Find the variance of X, the number of people who pick their own hat.

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Solution:

- We have

$$X = X_1 + X_2 + \cdots + X_n,$$

where X_i takes value 1 if the ith person selects the right hat and takes 0 otherwise.

- Noting that X_i is Bernoulli with parameter $p = 1/n$, we obtain

$$\mathbf{E}[X_i] = \frac{1}{n}, \quad \mathrm{var}(X_i) = \frac{1}{n}\left(1 - \frac{1}{n}\right)$$

Example

- For $i \neq j$, we have

$$\begin{aligned}
\mathrm{cov}(X_i, X_j) =& \mathbf{E}[X_i X_j] - \mathbf{E}[X_i]\mathbf{E}[X_j] \\
&= \mathbf{P}(X_i = 1 \text{ and } X_j = 1) - \frac{1}{n} \cdot \frac{1}{n} \\
&= \frac{1}{n} \cdot \frac{1}{n-1} - \frac{1}{n^2} \\
&= \frac{1}{n^2(n-1)}
\end{aligned}$$

- Therefore,

$$\begin{aligned}
\mathrm{var}(X) &= \sum_{i=1}^{n} \mathrm{var}(X_i) + \sum_{\{(i,j) | i \neq j\}} \mathrm{cov}(X_i, X_j) \\
&= n \cdot \frac{1}{n}\left(1 - \frac{1}{n}\right) + n(n-1) \cdot \frac{1}{n^2(n-1)} \\
&= 1.
\end{aligned}$$

Correlation coefficient

Definition of Correlation Coefficient

The **correlation coefficient** $\rho(X, Y)$ of two RVs X and Y that have nonzero variance is defined as

$$\rho = \mathbf{E}\left[\frac{(X - \mathbf{E}[X])}{\sigma_X} \cdot \frac{(Y - \mathbf{E}[Y])}{\sigma_Y}\right]$$
$$= \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

- $-1 \leq \rho \leq 1$
- $|\rho| = 1 \;\Leftrightarrow\; (X - \mathbf{E}[X]) = c(Y - \mathbf{E}[Y])$ (linearly related)
- Independent $\Longrightarrow$ $\rho = 0$ (**converse is not true**)

Example

- Consider n independent tosses of a coin with probability of a head equal to p. Let X and Y be the numbers of heads and of tails. What is the correlation coefficient of X and Y ?

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Solution:

- We have $X + Y = n$ and also $\mathbf{E}[X] + \mathbf{E}[Y] = n$. Thus,

$$X - \mathbf{E}[X] = -(Y - \mathbf{E}[Y]).$$

Furthermore,

$$\mathrm{cov}(X, Y) = \mathbf{E}\big[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])\big] = -\mathrm{var}(X) = -\mathrm{var}(Y).$$

Hence,

$$\rho(X, Y) = \frac{-\mathrm{var}(X)}{\sqrt{\mathrm{var}(X)\mathrm{var}(X)}} = -1.$$

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1. Show that

$$\rho(aX + b, Y) = \rho(X, Y).$$

2. **Schwarz inequality.** Show that for any RVs X and Y, we have

$$(\mathbf{E}[XY])^2 \le \mathbf{E}[X^2]\mathbf{E}[Y^2].$$

3. Consider the correlation coefficient of two RVs X and Y that have positive variances. Show that:
   (a) $|\rho(X, Y)| \le 1$.
   (b) If $Y - \mathbf{E}[Y]$ is a positive multiple of $X - \mathbf{E}[X]$, then $\rho(X, Y) = 1$.
   (c) If $\rho(X, Y) = 1$, then, with probability 1, $Y - \mathbf{E}[Y]$ is a positive multiple of $X - \mathbf{E}[X]$.

Outline

Summary

- Convolution

$$p_W(w) = \sum_x p_X(x) p_Y(w - x)$$

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x)\, dx$$

- Covariance

$$\mathrm{cov}(X, Y) = \mathbf{E}\big[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])\big]$$

- Correlation

$$\rho = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Readings: Section 4.1, 4.2
- Assignment: Problem Set 9
- Next lecture: Conditional expected value and transforms